

LOAN RISK & LENDING PERFORMANCE ANALYSIS

Top 3 Data Insights

Dataset: 38,576 loans | Total Funded: \$435.8M | Net Loss: \$28.2M

Tools: SQL + Power BI

Insight 1 — Risk Finding

Grade B and C are the Silent Risk — Not Grade G

What Everyone Assumes

When you look at the charge-off rate chart by grade, Grade G stands out at 31.31% — the tallest bar. Most people looking at this data would say the biggest risk is Grade G borrowers.

What the Data Actually Shows

Grade G only has 313 loans in the entire portfolio. That high percentage translates to just 98 actual defaults, representing \$6.3M of funded loans.

Grades B and C tell a completely different story:

Segment	Volume & Rate	Actual Defaults
Grade B	11,674 loans at 11.50% CO	1,343 defaults
Grade C	7,904 loans at 16.02% CO	1,266 defaults
B + C Combined	\$218.2M funded	2,609 defaults — 48.9% of all defaults
Grade G (comparison)	313 loans · \$6.3M funded	98 defaults — less than 2% of total

Nearly 1 in every 2 defaults in this portfolio comes from Grades B and C. Grade G, despite its alarming rate, is a small fraction of the actual damage.

Why This Happens — Volume Multiplies the Rate

A 16% default rate applied to 7,900 loans is far more damaging than a 31% rate applied to 313. Real portfolio risk is rate multiplied by volume — not rate alone. Grade G scores high on rate. Grades B and C score high on actual loss.

Sub-grade detail makes this clearer: sub-grade B5 defaults at 13.01%, almost identical to sub-grade C1 at 15.17%. The grade label boundary is not where the actual risk boundary sits.

KEY TAKEAWAY

The portfolio's biggest loss is coming from the middle of the risk scale, not the extreme end. A 2 percentage point improvement in Grade B and C default rates would prevent approximately 390 defaults and protect around \$15M. Tightening underwriting at sub-grades B4, B5, C3, C4, and C5 is the highest-impact intervention available.

HOW THIS WAS FOUND

The charge-off rate chart on Page 1 shows Grade G as the tallest bar. Hovering the tooltip reveals loan counts: G = 313, B = 11,674. Once you multiply rate by volume, the picture reverses entirely. This finding requires both the chart and the tooltip — neither alone tells the full story.

Insight 2 — Hidden Risk

The Term Trap — 60-Month Loans Are a Hidden Time Bomb

The Finding

The portfolio offers two loan terms: 36 months and 60 months. The performance difference between them is significant:

Term	Average Loan Size	Default Rate
36-Month	\$9,670	10.71%
60-Month	\$15,738	22.34%
Difference	1.6x larger loan	2.09x higher default rate

A 60-month borrower is twice as likely to default — and when they do, the bank loses 60% more money per loan because the loan amounts are larger. These two effects compound each other.

The Part That Makes It Worse — All Open Loans Are 60-Month

There are 1,098 loans in the portfolio currently marked as "Current" — still active, not yet resolved. Filtering these by term on the dashboard reveals:

- Filter Current loans by 36-month term → 0 results
- Filter Current loans by 60-month term → 1,098 results

Every single open loan in the portfolio is on a 60-month term. Not one active loan is on a shorter term. This means the 22.34% default rate for 60-month loans is understated — it only counts loans that have already closed. The 1,098 still-active loans will add to that count when they resolve.

If those loans default at even the current base rate, that adds approximately 245 more defaults and around \$8.6M in additional losses — losses not yet visible in the numbers.

Why This Is Called the Term Trap

Borrowers who need longer repayment periods are often already financially stretched. The bank approves them for 60-month terms and larger loan amounts, which looks like flexibility — but the data shows it concentrates risk. The longer the term, the more likely a borrower defaults before completing repayment.

RECOMMENDATION

Restrict 60-month loans to Grade A and B borrowers only. Grade C and below should be capped at 36-month terms. A term-grade eligibility matrix would address this directly: Grade A-B gets both options, Grade C and below gets 36-month only. Revisit after the Current loans resolve to confirm the projection.

HOW THIS WAS FOUND

Page 2 shows the 36M vs 60M comparison tiles (10.71% vs 22.34%). The open-loan finding required drilling into Page 3 — filtering Loan Status = Current, then toggling between term filters. The 36-month filter returns zero records. That is the moment the risk becomes visible. It is not findable from the summary page alone.

Insight 3 — Cross Insight

Grade × Purpose — Grade Drives Risk, But Business Breaks Every Grade

The Starting Question

Does the purpose of a loan tell you how risky it is — or does the borrower's credit grade matter more? This insight required crossing two dimensions of the data simultaneously, which is why it is invisible from either chart alone.

Finding 1 — Grade Dominates Purpose

When you look at the same loan purpose across different grades, the default rate changes dramatically. Debt Management loans are a clear example:

Debt Management Loans	Default Rate	
Grade A borrowers	5.6%	Safest group
Grade C borrowers	15.0%	Mid range
Grade F borrowers	31.6%	High risk
Spread — Grade A to F	26 percentage points	Same loan purpose, completely different risk

A 26 percentage point spread for the exact same type of loan tells you clearly: grade is the stronger predictor. The purpose tells you what the money is for. The grade tells you whether the person is likely to pay it back.

Finding 2 — Business Purpose Breaks the Rule

Small business loans carry elevated risk at every grade level — including Grade A, which is the bank's safest classification:

Segment	Default Rate	Note
Grade A — All other purposes	5.0% average	Expected safe zone
Grade A — Business purpose	11.4%	2x higher than all other Grade A loans
Grade B — Business	24.8%	High risk
Grade D — Business	30.8%	Very high risk

Grade G — Business	43.5%	Highest combination in portfolio
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Even a Grade A small business borrower — the bank's best classification — defaults at more than double the rate of other Grade A borrowers. The business-purpose risk is not captured by the grade. The current grading model underweights the inherent risk in small business lending.

Finding 3 — The Largest Dollar Risk Is Hidden in a Common Combination

The most dangerous combination in dollar terms is not the most obvious one. It is not Grade G + Business (43.5% default rate but only 46 loans and \$0.7M). The largest actual dollar exposure sits here:

BIGGEST DOLLAR RISK IN THE PORTFOLIO

Grade E + Debt Management: 1,840 loans | \$31.8M funded | 24.8% default rate This is a very common loan purpose (debt consolidation) at a moderate-high risk grade (E) — at significant scale. You cannot see this from either the grade chart or the purpose chart in isolation. It only appears when you cross-filter both dimensions together.

Recommendations

- Do not approve Business purpose loans at Grade D and below. The data shows 30.8%+ default rates — the grade alone is not compensating for the purpose risk.
- Monitor Grade E + Debt Management as a specific segment. With \$31.8M funded at a 24.8% charge-off rate, this is the single largest dollar risk concentration in the portfolio.
- Grade A borrowers remain the safest group across all purposes — with the one clear exception of small business, which should be treated as a separate underwriting category regardless of grade.

HOW THIS WAS FOUND

The Grade E + Debt Management finding required cross-filtering on the detail page (Page 3) — selecting Grade E from the grade slicer, then checking the purpose breakdown. The 1,840-loan, \$31.8M figure is invisible from either the grade or purpose chart alone. It only appears when both filters are applied simultaneously.

Project Note

All 3 insights are derived from the 38,576-row 2021 loan dataset. Every number is verifiable through the Power BI dashboard. No external benchmarks or assumptions were used.

Tools used: SQL (data extraction and aggregation) | Power BI (visualization and cross-filtering)